

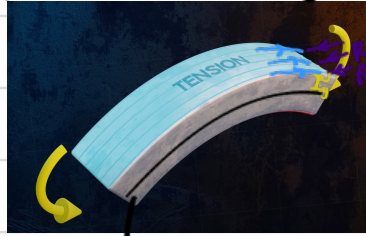
Type of Stress	Definition	Formula	Direction of Force	Example	Physical Effect When Yield Stress is Exceeded
Normal Stress	Stress acting perpendicular to the surface.	$\sigma = F/A$	Perpendicular to the surface	Pulling or compressing a material (e.g., stretching a rope or compressing a spring)	Tensile Stress: Material stretches (plastic elongation) or Compressive Stress: Material shortens or crushes, leading to permanent deformation. Axially Plastically Deformed
Axial Stress	A specific type of normal stress acting along the axis of a material (along its length).	$\sigma_{\text{axial}} = F/A$	Along the length (longitudinal axis)	A rod being stretched or compressed (e.g., a column under load)	Tensile Axial Stress: Material elongates beyond its elastic limit. Compressive Axial Stress: Material buckles or shortens beyond its original shape, leading to plastic deformation or instability. Axially Plastically Deformed
Shear Stress [NIS]	Stress that causes sliding or shearing between adjacent layers of a material.	$\tau = F/A$	Parallel to the surface	Cutting with scissors, shearing a material along a surface	Material slides or shears at the surface, causing permanent deformation (shearing failure). Shearly Plastically Deformed
Bending Stress	Stress experienced by a material subjected to a bending moment , causing one side to tension and the other side to compression .	$\sigma = M \cdot y/I$	Varies along the beam (tensile on one side, compressive on the other)	A beam bending under load, such as in a bridge or table	Tensile Side: Material elongates permanently (plastic stretching). Compressive Side: Material compresses permanently or buckles, causing plastic bending of the beam. Plastically Bent

Bending Stress, σ_B

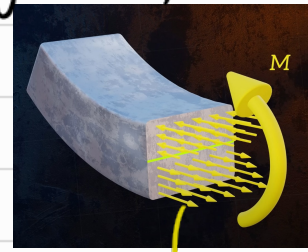
Internal Bending Stress, σ_B is caused by Bending Moment, M_B



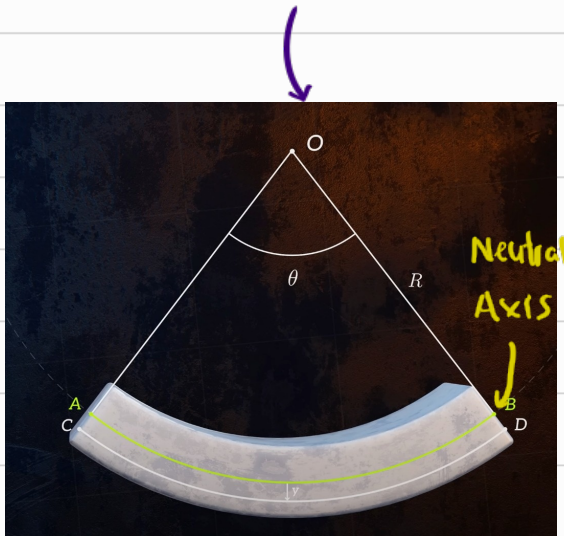
the axis with no stress
[Neutral Axis]



Neutral axis



Bending Stress



$$\text{Bending Strain, } \epsilon_B = \frac{\Delta L}{L} = \frac{CD - AB}{AB} = \frac{(R+y)\theta - R\theta}{R\theta} = \frac{y}{R}$$

$$\boxed{\epsilon_B = \frac{y}{R}}$$

$$E = \frac{\sigma}{\epsilon} \Rightarrow \text{Bending stress, } \sigma_B = \frac{E y}{R}$$

$$\text{Bending Moment, } M_B = \int \sigma_B y dA = \int \left(\frac{E y}{R} \right) y dA = \frac{E}{R} \int y^2 dA$$

$$\boxed{\sigma_B = E y K}$$

Curvature, $K = \frac{1}{R}$

$$\therefore I_B = \int y^2 dA$$

$$\boxed{M_B = \frac{E}{R} (I)}$$

Bending Inertia
[2nd moment of area]

$$M_B y = \left(\frac{E y}{R} \right) I$$

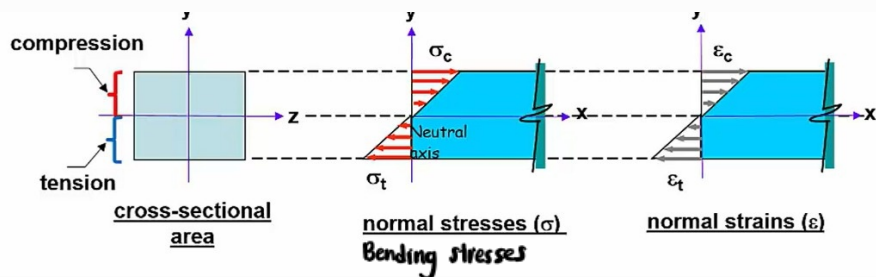
$$\boxed{\sigma_B = \frac{(M_B) y}{I}} = M/z, \text{ where } z = I/y$$

$$\boxed{M_B = E I K}$$

Curvature, $K = \frac{1}{R}$

As $R \uparrow$, curvature $\downarrow$, $\frac{1}{R} \downarrow$, $K \downarrow$

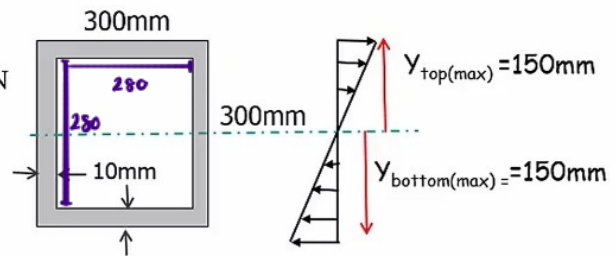
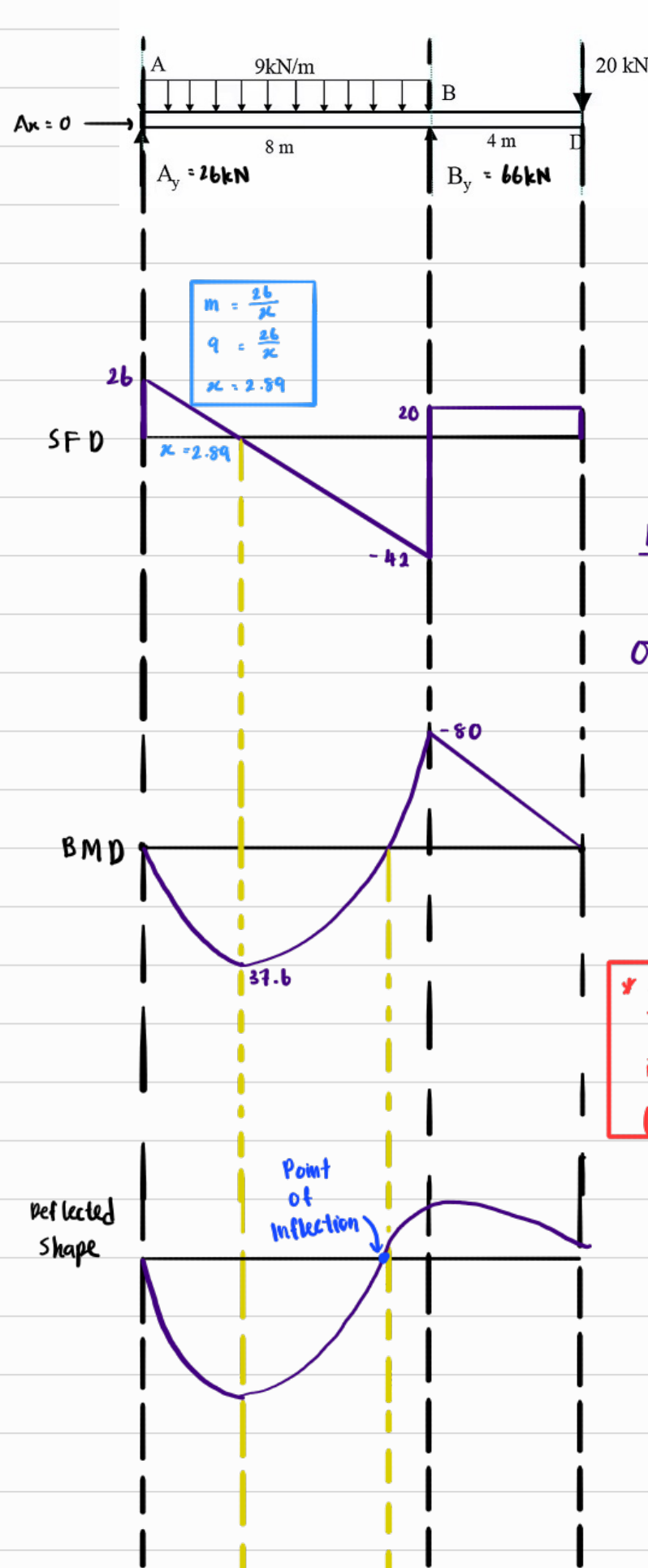
Distribution of Bending Stress



Since the fibers get more compressed and more stretched further from the neutral axis, the internal bending stress also increases away from the neutral axis.

Example 1:

A bridge, made of a steel RHS (rectangular hollow section), is used to support the loading shown. Is the strength of the steel bridge adequate? Assume $\sigma_y = 250 \text{ MPa}$



$$\begin{aligned}
 I &= \left(\frac{wh^3}{12} \right)_{\text{out}} - \left(\frac{wh^3}{12} \right)_{\text{in}} \\
 &= \frac{300(300)^3}{12} - \frac{280(280)^3}{12} \\
 &= (162.8 \times 10^6) \text{ mm}^4
 \end{aligned}$$

[from graph]

$$M_{B \text{ max internal}} = 80 \text{ when } y = \frac{300}{2} = 150 \text{ mm}$$

$$\sigma_{B \text{ actual max}} = \frac{(M_B)y}{I} = \frac{(80 \text{ kN})(150 \text{ mm})}{(162.8 \times 10^6) \text{ mm}^4} = 73.7 \text{ MPa}$$

$\therefore$ Since $\sigma_{B \text{ actual max}} < \sigma_{\text{yield}}$,

our beam is **adequate**

Acceptable in quality

* If there were external axial forces that causes internal axial stress, σ_{axial} , the sum of $(\sigma_{\text{axial}} + \sigma_{\text{bending}}) < \sigma_{\text{yield}}$.

Finding the Centroid axis (Neutral Axis) of Asymmetrical Shapes

Recall that when we find Centre of Mass, we essentially just replaced all moments acting at diff points to 1 equivalent moment acting at a single point.



Similarly, to find the Centroid, we essentially replace all 1st moment of area acting at diff points to 1 equivalent 1st moment of area acting at a single point.

$$[A_{total}] [y_{neutral}] = \sum A_i y_i$$

$$y_{neutral} = \frac{\sum A_i y_i}{A_{total}}$$

Centre of mass

Consider a rod of length 2m and mass 5kg.
We add a mass of 3kg to one end



If I want to represent the total moments of this system as just 1 force at a single point, the calculation will be as follows

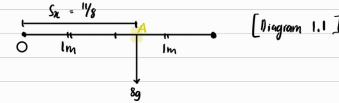
$$\begin{aligned} \sum \tau &= \sum \tau \\ \sum W_i x_i &= (\sum W) (x) \\ (5g)(1) + (3g)(2) &= (5g + 3g)(x) \\ (5 + 6)g &= (8g)(x) \\ x &= \frac{11}{8} \text{ m} \end{aligned}$$

Generic formula

$$x_c = \frac{\sum W_i x_i}{\sum W}$$

Dist. of C.O.M from point of reference

This means Diagram 1.0 can be represented by



* Key Terms

Centre of mass: The point where all weight act upon
[Balance point of area]

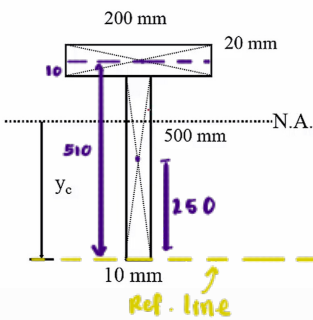
Centroid: Geometrical Centre of a body.

[Independent of mass distribution]

1st moment of area: Amount of leverage an area has to produce a moment about an axis

Example 1:

Find the neutral axis position.



$$\begin{aligned} y_{neutral} &= \frac{(200 \times 20)(510) + (500 \times 10)(150)}{[(200 \times 20) + (500 \times 10)]} \\ &= 365.5 \text{ mm} \end{aligned}$$

Finding 2nd moment of Area for Asymmetrical Shapes

Recall the parallel axis Theorem for rotational Inertia

Inertia is dependent on mass distribution, thus the term " Md^2 ".

[Area moment of inertia]

For 2nd moment of Area, it's a geometrical property that is dependent on area, independent of mass distribution. Thus the term " Md^2 " is replaced w " Ad^2 "

Parallel axis theorem [Reference point : CM]

Let the new axis, r'_i be d away from centre of mass [CM]

$$r'_i = r_i + d$$

So the total rotational inertia is,

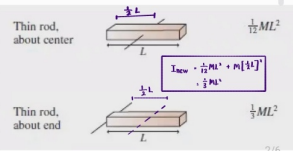
$$\begin{aligned} I_{\text{new}} &= \sum m_i [r_i + d]^2 \\ &= \sum m_i [r_i^2 + 2r_i d + d^2] \\ &= \sum m_i r_i^2 + 2d \sum m_i r_i + d^2 \sum m_i \end{aligned}$$

$$= I_{\text{CM}} + 0 + d^2 M$$

$$I_{\text{new}} = I_{\text{CM}} + Md^2$$

[CM]
Dist. b/w old axis & new axis

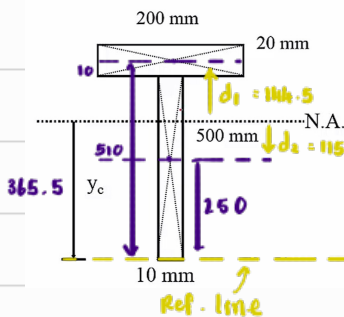
Since our ref. point is at CM, $\sum m_i r_i$ at CM = 0 hence all moments cancel out. That's why we can balance things when we put our finger at the CM



$$I_{\text{area new}} = \sum I_{\text{area individual}} + \sum A_i d_i^2$$

Dist from centroid to neutral axis

Example 1b)

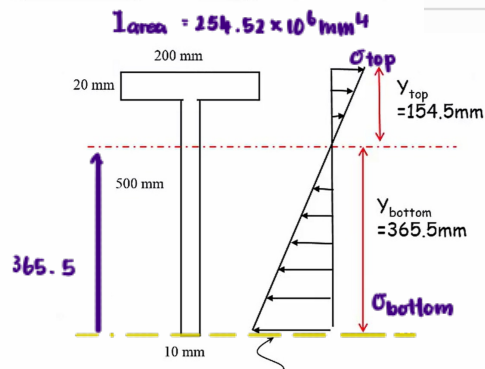
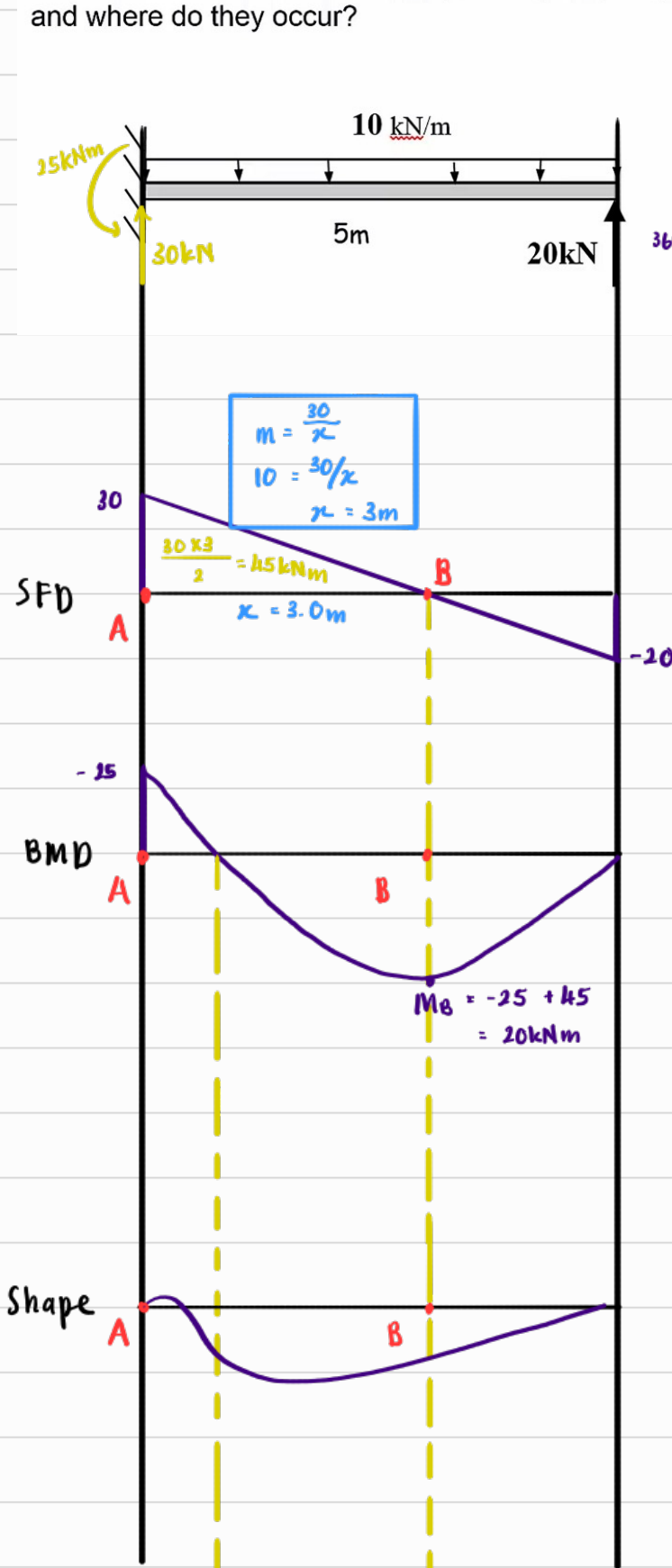


[Shown by Ques]

$$\begin{aligned} I_{\text{area new}} &= \left[\frac{(200)(20)^3}{12} + \frac{(10)(500)^3}{12} \right] + \left[(200 \times 20)(144.5)^2 + (10 \times 500)(115.5)^2 \right] \\ &= 254,572,250 \text{ mm}^4 \end{aligned}$$

Example 1 c)

A cantilever beam with the T-section shown, supports a UDL and a point load at its tip. It is made of cast iron, which is much stronger in compression than in tension. What are the maximum tension and compression bending stresses and where do they occur?



At point A

$$\sigma_{\text{bending bottom}} = \frac{(M_B)(y)}{I} = \frac{(25 \text{ kN})(365.5 \text{ mm})}{(254.52 \times 10^6 \text{ mm}^4)} = 26.1 \text{ MPa}$$

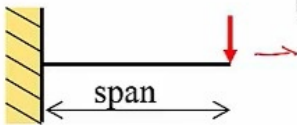
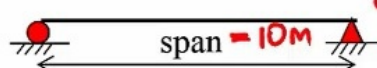
$$\sigma_{\text{bending top}} = \frac{(25 \text{ kN})(154.5 \text{ mm})}{(254.52 \times 10^6 \text{ mm}^4)} = 15.2 \text{ MPa}$$

At point B

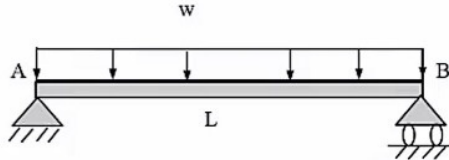
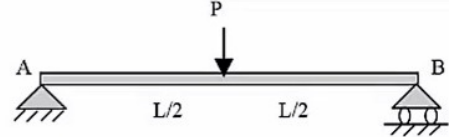
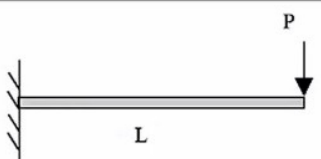
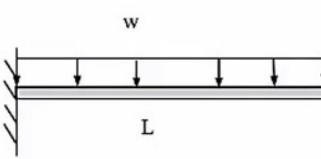
$$\sigma_{\text{bending top}} = \frac{(20 \text{ kN})(154.5 \text{ mm})}{(254.52 \times 10^6 \text{ mm}^4)} = 12.1 \text{ MPa}$$

$$\sigma_{\text{bending bottom}} = \frac{(20 \text{ kN})(365.5 \text{ mm})}{(254.52 \times 10^6 \text{ mm}^4)} = 28.7 \text{ MPa}$$

TWO LIMIT STATE CASES (design checks)

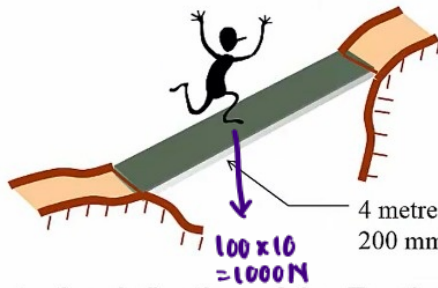
Ultimate Limit State (ULS)	Serviceability Limit State (SLS)
Design against Collapse	Design against excessive deflection
Load = 1.2DL + 1.5LL	Load = DL + 0.7LL ✖
Acting stresses < Allowable stresses $\sigma_{tension} (= \frac{P}{A}) < \sigma_{yield}$ $\sigma_{bend} (= \frac{My}{I}) < \sigma_{yield}$ $F_{comp} < \begin{cases} P_{euler} \\ \text{and} \\ \sigma_{yield} * A \end{cases}$	Maximum Deflection < $\begin{cases} \frac{span}{250} \\ \text{for beams} \\ \frac{span}{125} \\ \text{for cantilevers} \end{cases}$   defl < $\frac{10,000}{250}$ < 40mm

Maximum deflection formulae for beams

	$y_{max} = -\frac{5wL^4}{384EI}$
	$y_{max} = -\frac{PL^3}{48EI}$
	$y_{max} = -\frac{PL^3}{3EI}$
	$y_{max} = -\frac{wL^4}{8EI}$

NOTE: downwards deflection is negative

Example 1:



Loading = 100kg man at midspan.

Assume bridge to be weightless.

$E_{\text{timber}} = 10,000 \text{ MPa}$ *Dead Load (DL) = 0*

4 metre span footbridge.
200 mm x 40 mm timber plank.

Q. Calculate the deflection of the Footbridge at mid-span and check if the bridge meets the serviceability limit state requirements (deflect < span/250)

$$y_{\text{max}} = -\frac{PL^3}{48EI}$$

$$I = \frac{(200)(40)^3}{12} = 1.067 \times 10^6 \text{ mm}^4$$

$$\begin{aligned} \text{With SLS, Load} &= DL + 0.7LL \\ &= 0 + 0.7(1000) \\ &= 700 \text{ N} \end{aligned}$$

$$\begin{aligned} y_{\text{max}} &= -\frac{(700 \text{ N})(4000 \text{ mm})^3}{48(10,000 \text{ MPa})(1.067 \times 10^6 \text{ mm}^4)} \\ &= 87.5 \text{ mm} \quad [\text{Max deflection}] \end{aligned}$$

$$\text{Allowable deflection} = \frac{\text{span}}{250} = \frac{4000}{250} = 16 \text{ mm}$$

$\therefore$ Since $87.5 > 16$, the bridge fails SLS conditions.